Advance Data Science

Module 1: Statistical Inference – Foundations & Distributions

Topic: Probability Distributions Applied

- Normal, binomial, and Poisson distributions
- Sampling distributions, standard error, and Central Limit Theorem (CLT)
- Confidence intervals: construction and interpretation

Module 2: Statistical Inference – Hypothesis Testing

Topic: Hypothesis Testing Principles

- Null/alternative hypothesis, Type I/II errors, and p-value interpretation
- One-sample and two-sample t-tests with worked examples
- Chi-square test for independence and goodness of fit
- ANOVA for comparing group means

Module 3: Ensemble Methods – Boosting Algorithms

Topic: Sequential Boosting Techniques

- **AdaBoost:** sequential weighting, weak learners, and final classifier construction
- **Gradient Boosting:** residual fitting, shrinkage, and tree depth optimization
- **XGBoost:** regularization, parallel tree building, and missing value handling
- **LightGBM:** leaf-wise growth, categorical features, and speed advantages

Module 4: Ensemble Methods – Stacking & Voting

Topic: Model Aggregation & Ensembling

- **Voting Classifiers:** hard voting, soft voting, and operational use cases
- **Stacking:** base learners, meta-learner, and out-of-fold predictions
- **Blending vs. Stacking:** architectural differences and trade-offs

Module 5: Natural Language Processing – Foundations

Topic: NLP Fundamentals & Text Representations

- Text preprocessing and tokenization pipelines
- TF-IDF (Term Frequency–Inverse Document Frequency)

- Word embeddings and contextual embeddings

Module 6: Natural Language Processing – Applications & Systems

Topic: Applied NLP & Conversational AI

- Text classification and sentiment analysis models
- Information extraction and Named Entity Recognition (NER)
- Conversational AI and chatbot development: architectural overview and design patterns

Operating Systems and Security

Module 1: Introduction to Operating Systems

Topic: OS Fundamentals

- What an Operating System does
- Kernel
- System calls
- Types of Operating Systems: Batch, Time-Sharing, Real-Time, and Distributed
- Operating System as a resource manager: CPU, Memory, and I/O

Module 2: Process Management

Topic: Processes

- Process concept and the process life cycle
- Process states: New, Ready, Running, Waiting, and Terminated
- Process Control Block (PCB)
- Context switching

Topic: CPU Scheduling

- First Come First Serve (FCFS)
- Shortest Job First (SJF)
- Priority Scheduling
- Round Robin Scheduling
- Comparison of scheduling algorithms: Throughput, Turnaround Time, and Waiting Time

Module 3: Memory & I/O Management

Topic: Memory

- Memory hierarchy
- Memory allocation
- Paging
- Segmentation
- Virtual memory

Demand paging

Deep Learning and Generative Architectures

Module 1: Differentiation & Differential Equation

Topic: Differentiation

- Differentiation
- Rules of Differentiation (Product Rule, Quotient Rule, Chain Rule)
- Partial Derivatives
- Higher Order Derivative

Module 2: Integration & Vector Calculus

Topic: Integration & Vector Calculus

- Integration
- Definite and Indefinite Integration
- Area under the curve
- Multiple integration (basics)
- Basic Vector
- Vector Field
- Curl

Module 3: Biological Neuron Analogy

Topic: ML vs DL vs AI vs GenAI

- ML vs DL vs AI vs GenAI
- Notable differences
- Mapping human brain to neurons/neural networks

Module 4: Neural Network Refresher

Topic: Perceptron to MLP (Concept)

- Perceptron, MLP, activation functions (ReLU, LeakyReLU, sigmoid, tanh, GELU, Swish,Softmax)
- Forward pass: matrix multiplications, activations, output layer
- Loss functions: MSE for regression, cross-entropy for classification
- Backpropagation: chain rule through computational graph step-by-step
- Weight initialization: Xavier, He, why initialization matters

Topic: Regularization & Normalization Essentials

- Dropout, weight decay, batch/layer normalization, early stopping — essentials only

Module 5: Optimization Refresher

Topic: Loss Landscapes & Optimizers

- Learning rate schedules, saddle points/local minima intuition, gradient clipping; SGD/Adam/AdamW applied in PyTorch

Module 6: PyTorch Refresher

Topic: Tensors, Autograd & Training Loop (Hands-on)

- Tensors & autograd; nn.Module; DataLoader/Dataset; full training loop (forward, loss, backward, optimizer step) — hands-on refresher lab

Module 8: Tokenization & Embeddings

Topic: Subword Tokenization & Embeddings

- Subword Tokenization & Embeddings

Module 9: Attention Mechanism

Topic: Scaled Dot-Product & Multi-Head Attention

- Scaled Dot-Product & Multi-Head Attention

Module 10: Transformer Glass-Box Workshop

Topic: Build a Mini-GPT from Scratch

- Build a Mini-GPT from Scratch

Module 11: Autoregressive Generation

Topic: Causal Masking & Decoding Strategies

- Causal Masking & Decoding Strategies

Module 12: Landscape of Generative Architectures

Topic: Comparing Generative Paradigms

- Comparing Generative Paradigms

Module 13: LLM Foundations Survey

Topic: Different Ways of Prompting

- Different Ways of Prompting

Module 14: Prompt Engineering & Context Design

Topic: Principles & Structured Prompting

- Principles & Structured Prompting

Module 15: Data Curation & Synthetic Data

Topic: Curating Fine-Tuning Datasets

- Curating Fine-Tuning Datasets

Module 17: Evaluation-Driven Development

Topic: Eval Sets, LLM-as-Judge & Regression

- Eval Sets, LLM-as-Judge & Regression

Module 18: LLM APIs & Retrieval-Augmented Generation

Topic: APIs, Embeddings & RAG Pipeline

- APIs, Embeddings & RAG Pipeline

Module 19: AI Agents & Workflows

Topic: Tool-Calling, Planning & Multi-Agent Basics

- Tool-Calling, Planning & Multi-Agent Basics

Computer Vision & Multimodal AI

Module 1: Image Processing Foundations

Topic: Fundamentals

- Image representation, pixels, color spaces
- Image processing and visual representation learning
- Classical CV techniques and feature extraction

Module 2: Classical CV & Feature Engineering

Topic: Spatial Filtering

- Convolution-based filtering: blurring and sharpening kernels

Topic: Edge & Morphological Operations

- Edge detection (Sobel, Canny); morphological ops and histogram equalization

Topic: Feature Descriptors

- Keypoint detection and description: SIFT, HOG, feature matching

Topic: Motion Basics

- Optical flow fundamentals and frame-to-frame motion estimation

Topic: CV Core Challenges

- Viewpoint variation, illumination changes, occlusion, scale variation

Module 3: Core CV Tasks

Topic: CNN Architecture Mechanics

- How convolution works (sliding-window operation, kernel/stride/padding, output-size intuition)
- Pooling (max/average) - (practical implementation) and why it reduces dimensions

Topic: Vision Techniques

- Image classification with CNNs and pre-trained models
- Object detection (YOLO, R-CNN family)
- Image segmentation (semantic and instance)
- Image generation (GANs, diffusion overview)

Module 4: Building CV Applications

Topic: Applied CV

- Transfer learning with pre-trained models - Tour of the architecture lineage (LeNet->AlexNet->VGG->ResNet->EfficientNet) - what problem each innovation solved (e.g. skip connections vs. vanishing gradients)
- Using deep learning frameworks for CV

Module 5: Applied CV Lab Practice

Topic: Detection Lab

- Build and apply a face detector

Module 6: Generative Models

Topic: GANs

- GAN framework: generator, discriminator, minimax objective
- Training dynamics: Nash equilibrium, mode collapse, instability
- DCGAN & Conditional GAN: convolutional architecture and class-conditional generation

Problem Solving and Critical Thinking

- Problem Understanding & Structured Problem Solving
- Problem Decomposition & Root Cause Analysis
- Algorithmic Thinking
- Information & Computational Thinking
- Searching & Exploration
- Algorithmic Efficiency
- Computational Efficiency & Scalability